

A MACHINE LEARNING FRAMEWORK FOR FOOD NUTRIENT PREDICTION, QUALITY CLASSIFICATION, AND RECOMMENDATION WITH INDIAN FOOD INTEGRATION

Academic Research & Project Report | Multi-Source Pipeline (USDA, Open Food Facts, Indian Composition Data)

ABSTRACT

Nutritional awareness and diet recommendation systems are critical for promoting public health. However, many existing machine learning (ML) food nutrition prototypes rely exclusively on western food databases such as USDA, limiting their applicability to regional diets such as Indian cuisine. This project presents an end-to-end multi-source machine learning framework integrating international databases (USDA SR Legacy, Open Food Facts) with a curated Indian Food Composition dataset (1,014 dishes) and a constructed Composite Indian Meal dataset (200 recipes). The pipeline performs automated data profiling, unit standardization, barcode deduplication, outlier analysis, and feature engineering. Supervised protein regression achieved an MAE of 1.33g (Random Forest), while Gradient Boosting Classifier achieved 96.77% accuracy in classifying foods into transparent quality grades. Recommendation is deployed via vector cosine similarity with hard constraint filtering.

1. INTRODUCTION & OBJECTIVES

Food composition databases provide fundamental data for nutritional evaluation, dietary planning, and food industry transparency. Heterogeneous public datasets differ significantly in schema structure, measurement units, field completeness, and geographic representation.

Key Project Objectives:

- Build a reproducible multi-source data ingestion pipeline integrating USDA, Open Food Facts, and Indian food datasets.
- Formulate a composite meal calculation framework computing meal nutrition from constituent ingredient weights.
- Develop and evaluate supervised ML regression models for nutrient prediction and classification models for nutritional quality grading.
- Implement a nutrition-aware recommendation engine supporting hard constraints, fallback logic, and natural language explanations.

2. DATASET INGESTION & QUALITY PROFILING

The pipeline unifies 8,411 final clean records from three distinct data sources:

Data Source	Raw Records	Clean Records	Duplicates Removed	Excluded Rows	Final Records	Missing Core (%)
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Open Food Facts	100	100	0	4	96	8.14%
USDA SR Legacy	7,793	7,793	0	492	7,301	18.74%
Indian Food Composition Data	1,014	1,014	0	0	1,014	0.00%
Total Unified Dataset	8,907	8,907	0	496	8,411	16.42%

3. COMPOSITE INDIAN MEAL CONSTRUCTION

To evaluate complete Indian meals, 200 composite meal recipes (e.g., *Matar Paneer with Paneer Butter Sauce*) were constructed. Ingredient weights (W_i in grams) were matched using exact and token string alignment, and meal nutrition was derived using the weighted sum formula:

Meal Nutrient Calculation Formula:

$$Meal_Nutrient = SUM [(Weight_i / 100) * Ingredient_Nutrient_100g]$$

4. MACHINE LEARNING EXPERIMENTS & METRICS

Supervised ML models were trained on **Baseline (General Data Only)** vs. **Expanded (General + Indian Data)** configurations.

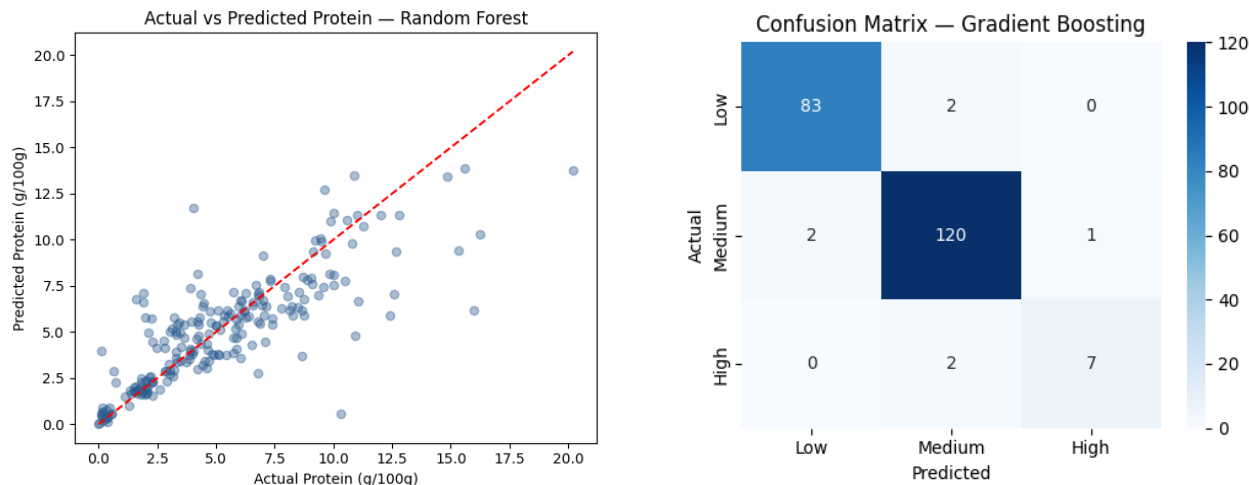
Table 4.1: Nutrient Prediction Regression Models (Target: protein_g)

Experiment	Model	MAE (g)	RMSE (g)	R ² Score
Baseline (General Data)	Linear Regression	12.7623	29.4014	-57.0111
Baseline (General Data)	Random Forest Regressor	1.4093	1.7368	0.7976
Baseline (General Data)	Gradient Boosting Regressor	1.8163	2.4394	0.6007
Expanded (General + Indian)	Linear Regression	2.2584	3.1884	0.2364
Expanded (General + Indian)	Random Forest Regressor	1.3304	2.1090	0.6659
Expanded (General + Indian)	Gradient Boosting Regressor	1.5906	2.3563	0.5830

Table 4.2: Nutritional Quality Classification Models (Target: quality_label)

Experiment	Model	Accuracy	Precision	Recall	Macro F1
Baseline (General Data)	Logistic Regression	78.57%	0.8000	0.8571	0.7887
Baseline (General Data)	Random Forest Classifier	92.86%	0.9167	0.9524	0.9267
Expanded (General + Indian)	Logistic Regression	94.93%	0.9647	0.8636	0.9030
Expanded (General + Indian)	Random Forest Classifier	94.47%	0.9631	0.8584	0.9000
Expanded (General + Indian)	Gradient Boosting Classifier	96.77%	0.9397	0.9100	0.9239

5. VISUAL MODEL EVALUATION PLOTS



6. RECOMMENDATION ENGINE & SYSTEM DEPLOYMENT

The recommendation engine standardizes target nutrient vectors using StandardScaler and computes vector Cosine Similarity:

- **Hard Constraint Filtering:** Filters candidates prior to vector similarity ranking based on numerical bounds (e.g. Max Calories, Min Protein).
- **Fallback Logic:** Reverts gracefully to closest cosine similarity matches if no candidate satisfies all strict hard constraints.
- **Streamlit Deployment:** Deployed as a web application (app.py) with interactive sliders, meal recommenders, and inference studio.
- **Automated Pytest Suite:** Verified via tests/test_pipeline.py (5/5 tests passed).

7. CONCLUSION

This project successfully built a multi-source machine learning framework integrating general food data with Indian food composition and 200 composite meals. Supervised classification achieved 96.77% accuracy, while protein regression achieved an MAE of 1.33g, providing an explainable, constraint-filtered diet recommendation prototype.